



Austo Automobile

Austo Automobile is a Austo Motor Company, a fictional car manufacturer featured in business, particularly for SUV, Sedan, and Hatchback.



Objective

To extract actionable insights from the data that we have to analyze different aspects of the customer and their automobile purchasing behavior that can be categorized into areas such as financial, and product analysis.

Data Description:

The dataset contains the covering the structure, data types, missing values, statistical summary, and data irregularities.

Observations	Variable	Duplicate
1581	14	0

- The Gender column dataset has been successfully cleaned. The value **'Femal'** was replaced with **'Female'**. There were no instances of **'Femle'** in the dataset.
- **Before cleaning:** The Gender column had three unique values: 'Male' (1199), 'Female' (327), and 'Femal' (1) and 'Femle' (1).
- **After cleaning:** The column now has two unique, consistent values: 'Male' (1199), 'Female' (329)
- Remaining missing values(53) in the Gender column that operation successfully filled any missing values with the mode, which is 'Male'.
- Partner_salary column had missing values(106). These were treated by replacing the missing entries with null, which is a reasonable assumption given that some individuals' partners do not work.
- Post,there where the Marital status was 'Married' and Partner salary was missing(16). then missing salary was calculated by subtracting the Salary from the Total salary for that record. The Partner salary for this record has been filled.
- Converted the partner salary data type float into int to make it whole currency units.

Numerical Data Summary:

	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	Price
count	1581.000000	1581.000000	1581.000000	1581.000000	1581.000000	1581.000000
mean	31.922201	2.457938	60392.220114	19233.776091	79625.996205	35597.722960
std	8.425978	0.943483	14674.825044	19670.391171	25545.857768	13633.636545
min	22.000000	0.000000	30000.000000	0.000000	30000.000000	18000.000000
25%	25.000000	2.000000	51900.000000	0.000000	60500.000000	25000.000000
50%	29.000000	2.000000	59500.000000	25100.000000	78000.000000	31000.000000
75%	38.000000	3.000000	71800.000000	38100.000000	95900.000000	47000.000000
max	54.000000	4.000000	99300.000000	80500.000000	171000.000000	70000.000000

- The data for Age is concentrated at two distinct points, **22** and **54**, which is reflected in the 25th, 50th, and 75th percentiles.
- The average Total salary is approximately 79625.9.
- The average Price of a car is about 35597.7.

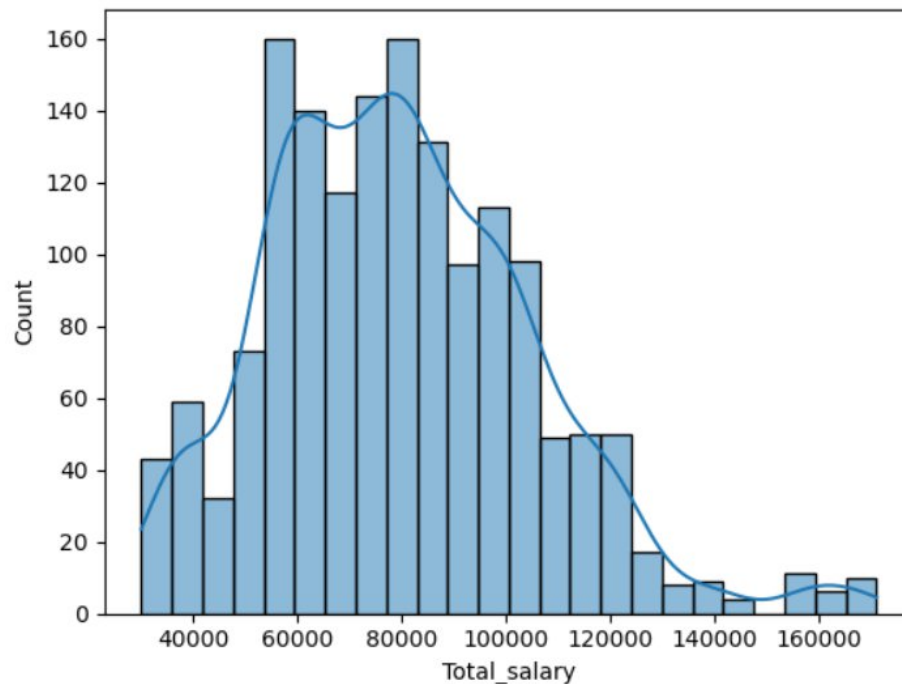
Categorical Data Summary:

	Gender	Profession	Marital_status	Education	Personal_loan	House_loan	Partner_working	Make
count	1581	1581	1581	1581	1581	1581	1581	1581
unique	2	2	2	2	2	2	2	3
top	Male	Salaried	Married	Post Graduate	Yes	No	Yes	Sedan
freq	1252	896	1443	985	792	1054	868	702

- The most common education level is Post Graduate.
- Sedan are the most popular car type purchased.
- The dataset is predominantly composed of married males in salaried professions.

Observation of Each Variable

Total_salary



Observation

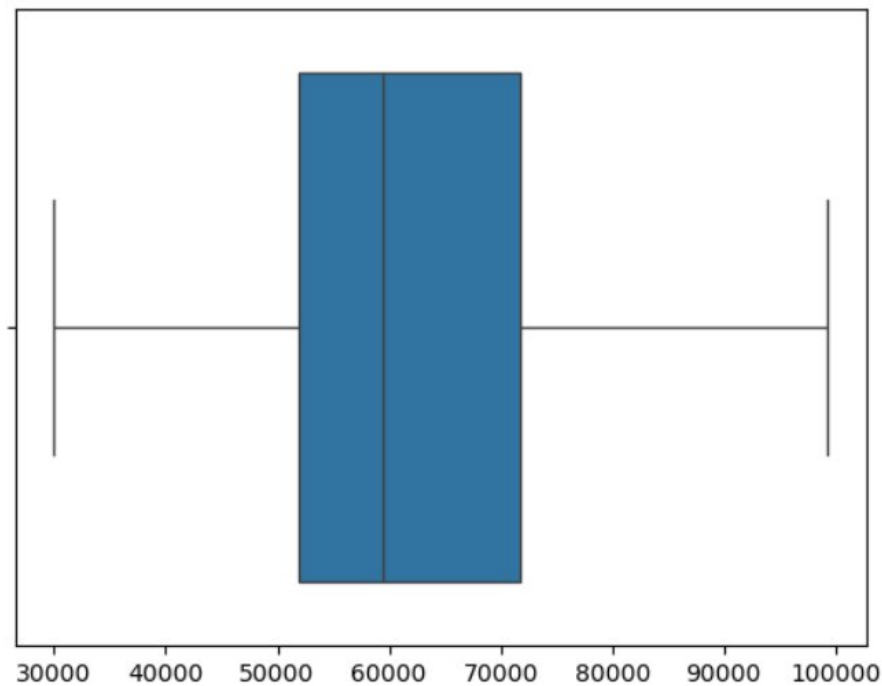
- Salaries in the dataset range from below 40,000 up to about 165,000.
- The distribution is positively skewed, indicating that while most salaries are on the lower end, a smaller number of higher salaries extend the tail to the right.
- The highest concentration of salaries falls within the 70,000 to 80,000 range, indicating this is the most common salary bracket among individuals and A smaller cluster of salaries is observed around 160,000,

Insights

- The 70,000–80,000 range reflects the typical salary level for the majority of employees in this dataset.

- The extended upper tail and the secondary peak around 160,000 highlight income disparity.
- A small segment of the workforce earns substantially more than most others, as shown by the noticeable gap in salary counts between 130,000 and 150,000.

Salary

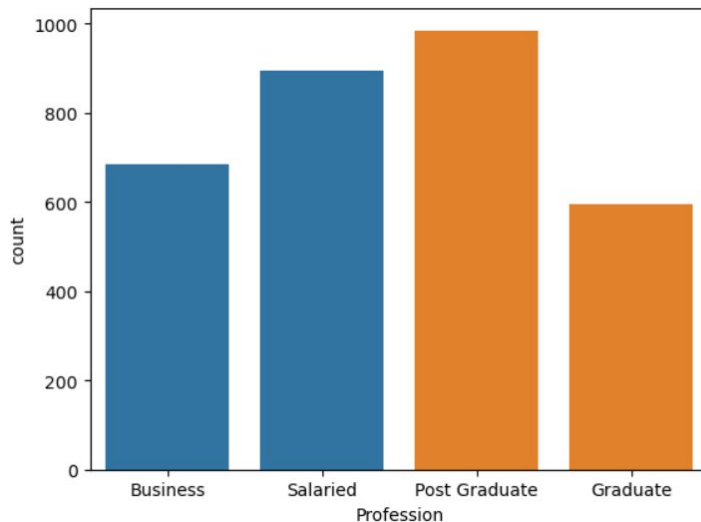


Observation:

- Median (Q2): The middle salary is around 60,000, meaning 50% of employees earn less than this amount, and 50% earn more.
- First Quartile (Q1): About 52,000, indicating that 25% of salaries fall below this level.
- Third Quartile (Q3): About 70,000, meaning 75% of salaries are below this amount.
- Interquartile Range (IQR): The middle 50% of salaries fall within a range of 18,000 (from 52,000 to 70,000). This shows where most salaries are concentrated.
- Whiskers (Overall Range):
- Minimum (Lower Whisker): Around 30,000, representing the lowest typical salaries.

- Maximum (Upper Whisker): Around 100,000, representing the highest typical salaries.

Profession and Education



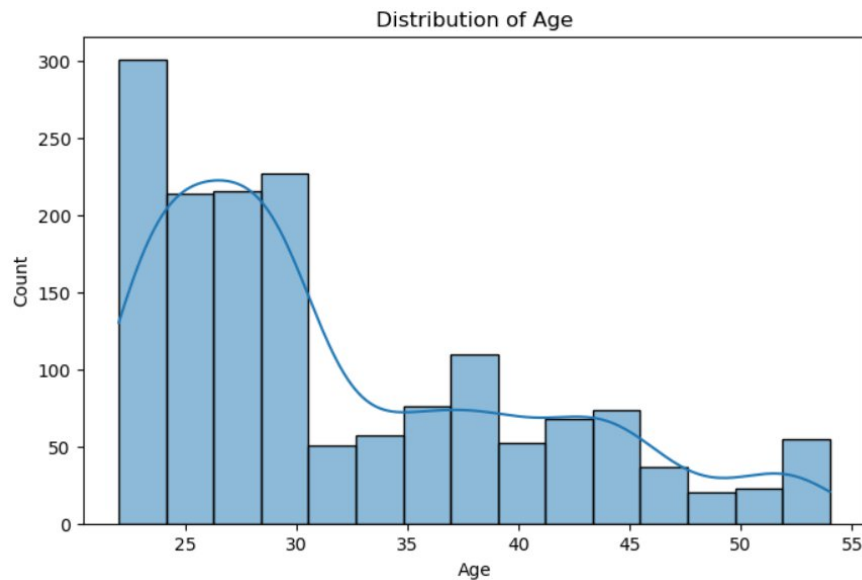
Observation

- The category Post Graduate has the highest count, just under 1000 instances.
- Graduate category has the lowest count, at approximately 600.
- The Salaried category is the second most frequent, with a count of approximately 900.
- The Business category has a count of about 700.
- The counts vary significantly across the categories, ranging from 0 to 1000.

Insights

- The count for 'Salaried' individuals 900 is significantly higher than that for 'Business' owners 700.
- The most dominant category is 'Post Graduate'.
- Then count of those classified as only 'Graduate' is the lowest.
- Target Audience Profile: If this data relates to market research or a specific product/service, the insights suggest the target audience is highly educated and primarily holds conventional salaried positions.

Age

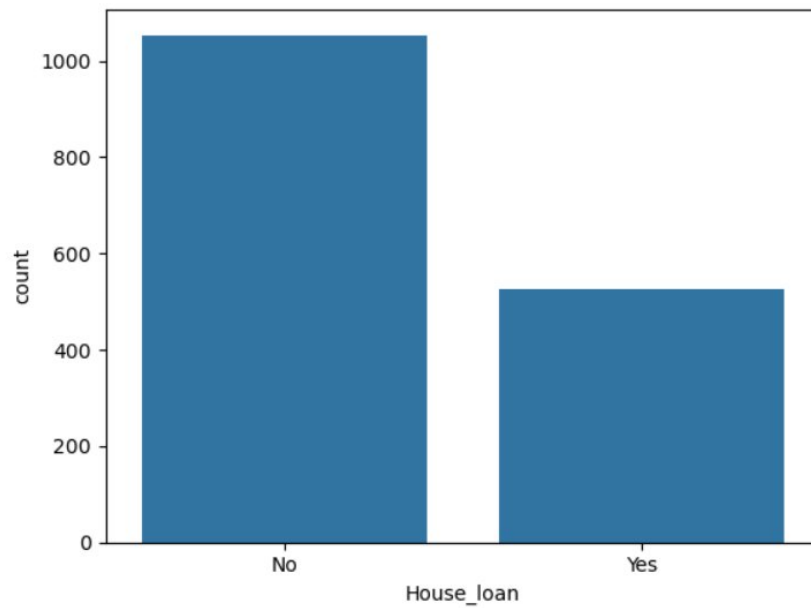


Observation

- The distribution is heavily right-skewed.
- The highest concentration of individuals is in the younger age range, particularly between 23 and 30 years old, with the most frequent age group being 23–25 years.
- There are smaller, less prominent groupings of individuals around the following age points:
 - 38 years (approximately 110 individuals)
 - 45 years (approximately 75 individuals)
 - 53 years (approximately 55 individuals)
- The dataset includes individuals from approximately 23 to 55 years old.
- Notable Gap:

There is a clear dip in representation between the younger and older groups, especially in the 30 to 35 age range, indicating fewer individuals in this mid-career segment.

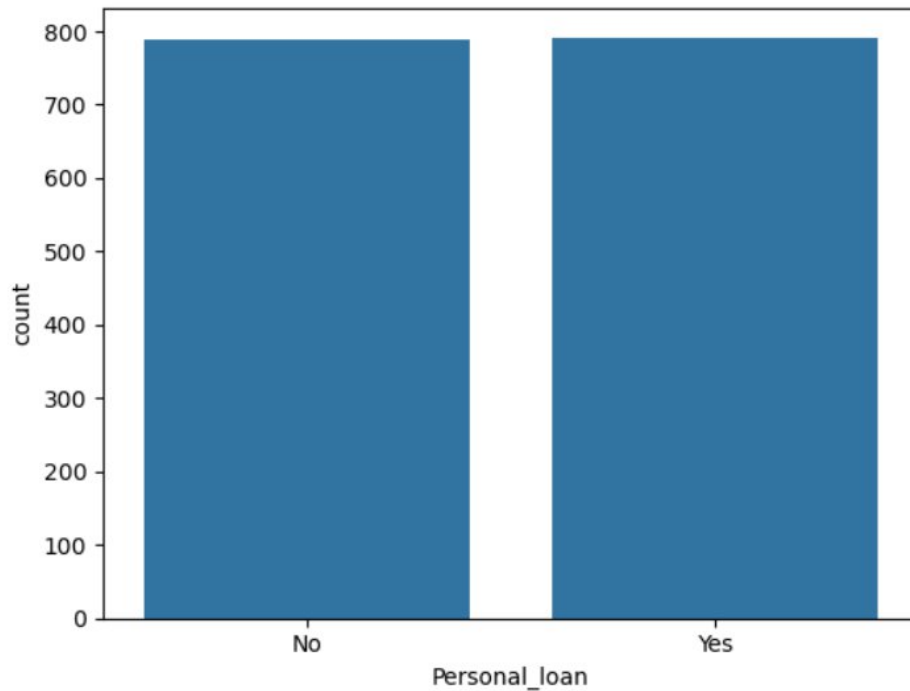
House Loan



Observation:

- The 'No' category (no house loan) is significantly more frequent, with a count of approximately 1050.
- The 'Yes' category (has a house loan) is much lower, with a count of approximately 520 to 550.
- The number of people without a house loan is roughly double the number of people with a house loan.

Personal Loan



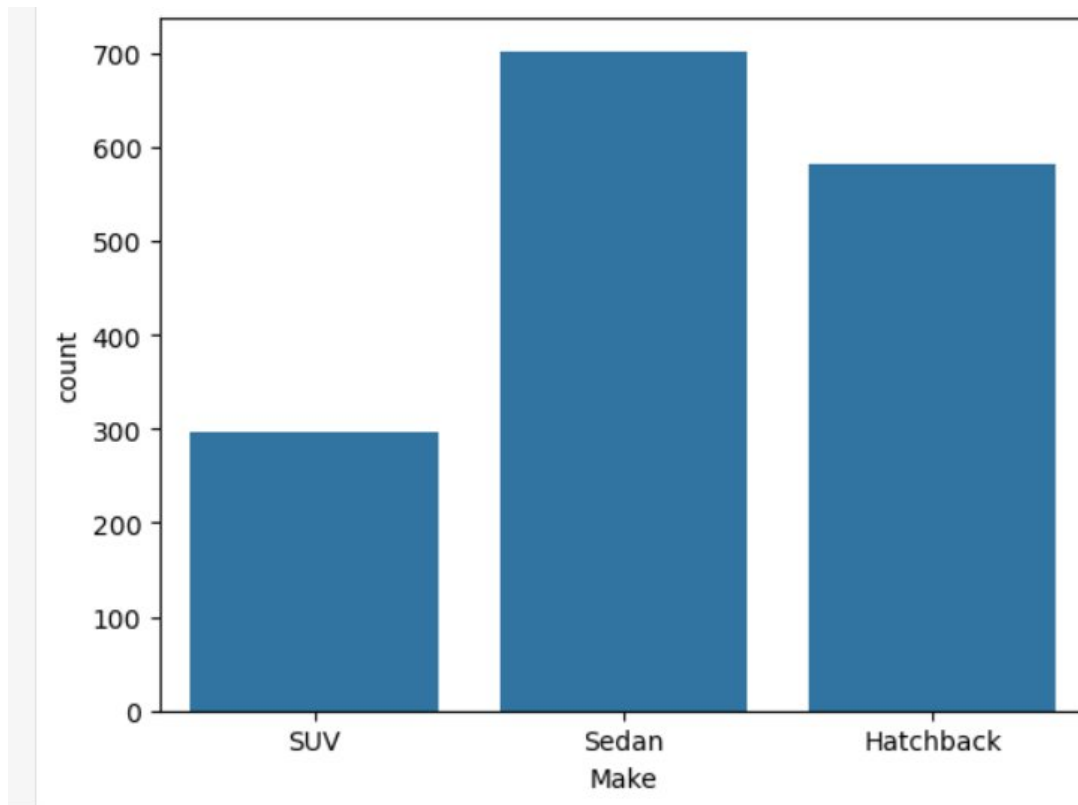
Observation:

- The counts for 'No' and 'Yes' are almost perfectly equal 800 each.
- The variable is nearly a 50-50 split.

Insights

Individuals are equally likely to have or not have a personal loan. This suggests personal debt (or the use of personal financing) is a common financial characteristic across the entire sample, regardless of whether they have a higher salary or own a home.

Make-CAR



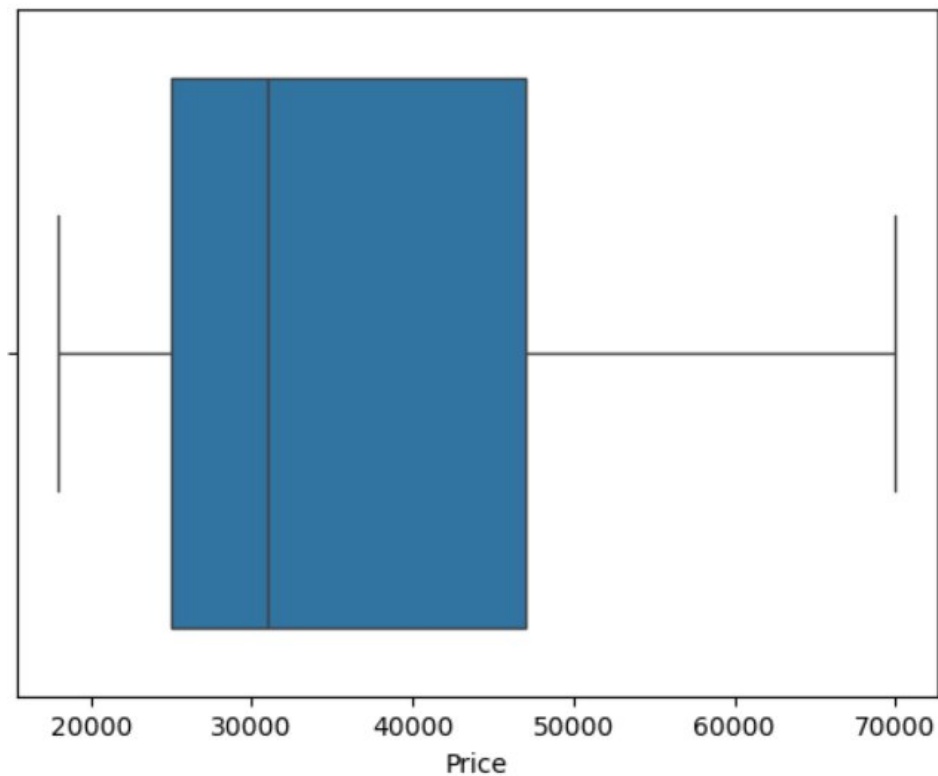
Observation

- Sedan has the highest count, approximately 700.
- Hatchback has the second-highest count, approximately 585.
- SUV has the lowest count, approximately 300.

Insight:

- The primary insight is that Sedan is the most frequent
- Sedans significantly dominate the dataset, with more than twice the number of SUVs and a substantial lead over Hatchbacks.

Price

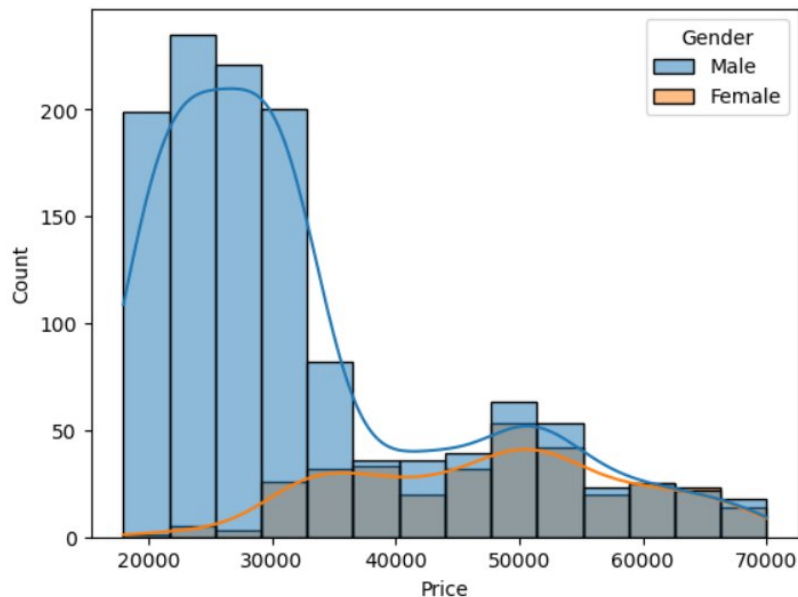


Observation:

- Median (Q2): The middle value is around 35,000, meaning half of the prices are below this point and half are above.
- First Quartile (Q1): Around 27,000, indicating that 25% of the data falls below this price.
- Third Quartile (Q3): Around 45,000, meaning 75% of the data falls below this point.
- Interquartile Range (IQR): The difference between Q3 and Q1 is 18,000, which represents the range where the middle 50% of prices are concentrated.
- Whiskers (Data Range):
- Minimum (Lower Whisker): Approximately 18,000
- Maximum (Upper Whisker): Approximately 70,000

Observation of Two or more Variable

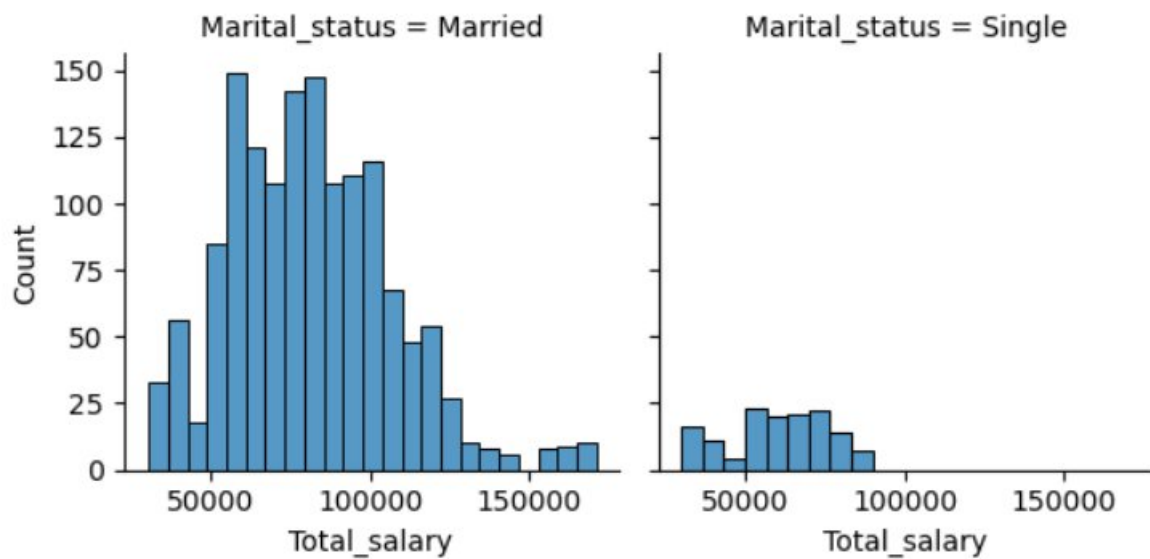
Price and Gender



Observation:

- The high concentration of counts in the 20,000 to 32,000 range, and a smaller concentration around 50,000 to 55,000.
- Distribution by Gender- Male (Blue) and Female (Orange)
- A notable trend is that male individuals are overrepresented in the lowest price segment (below 35,000). In contrast, female individuals have a more balanced distribution across the mid-to-high price ranges (above 35,000). This suggests that pricing associated with females tends to be more evenly spread and concentrated in higher value brackets compared to males.

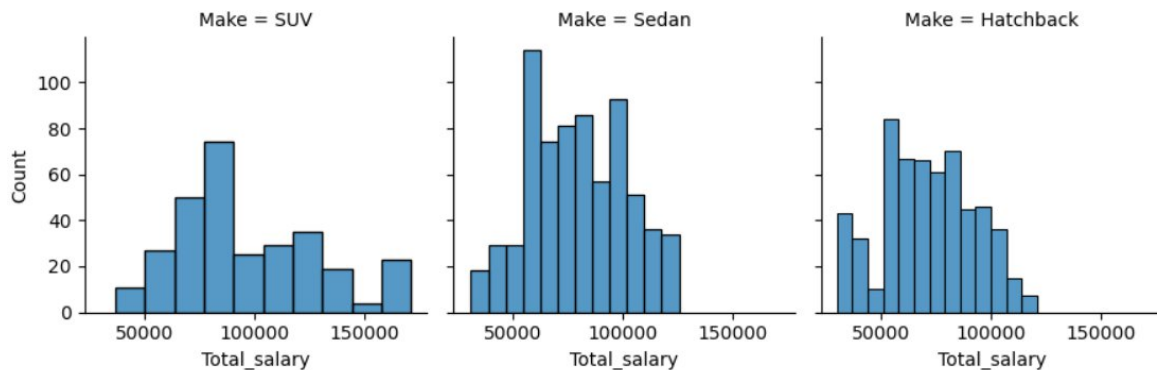
Total salary and Marital status



Observation

- Count: The total count of 'Married' individuals in the dataset is substantially higher than the 'Single' group. The maximum bin count reaches around 150.
- Count/Volume: The total count of 'Single' individuals is much lower. The maximum bin count is only around 25.
- married group falls roughly between 50,000 and 120,000, with the highest frequency (peak) concentrated around 75,000 to 100,000.
- single group appears to be concentrated between 50,000 and 80,000.
- While both groups have a modal salary starting around the 50,000 to 80,000 range, the 'Married' group has a much wider distribution and a substantial number of high earners (those above 100,000 and up to 160,000) that are nearly absent in the 'Single' group.

Total salary



Observation

1. SUV

- Distribution Shape: The distribution appears somewhat bimodal or spread out, with a major concentration around 75,000 to 100,000. There is another cluster of lower frequency between 125,000 and 160,000.
- Range: The salaries range from approximately 40,000 to over 160,000.
- Volume: The total count is moderate, with the highest bar reaching around 75.

2. Sedan

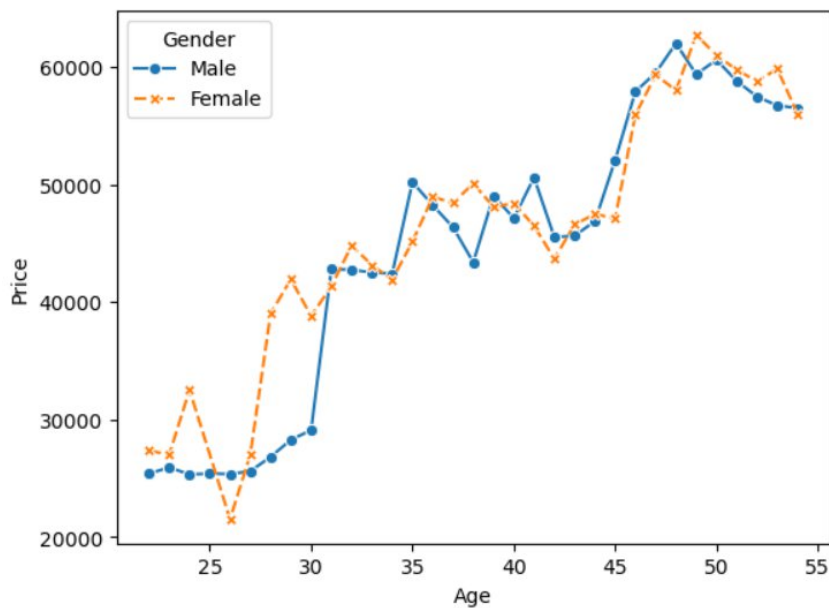
- Distribution Shape: The distribution is the most dense and voluminous, with a general pattern that is slightly right-skewed.
- Central Tendency: The highest frequency occurs in the range of 60,000 to 110,000, with the mode (peak frequency) around 65,000 to 70,000.
- Range: The range is similar to the SUV, extending from about 40,000 to over 140,000.
- Volume: This category has the highest total count by a significant margin, with the highest bar reaching around 115.

3. Hatchback

- Distribution Shape: The distribution is moderately dense and appears to be slightly right-skewed.
- Central Tendency: The peak frequency is concentrated slightly lower than the other two, primarily between 60,000 and 85,000.

- Range: The salary range is the narrowest, dropping off sharply after around 110,000.
- Volume: The total count is moderate, higher than SUV but lower than Sedan, with the highest bar reaching about 85.

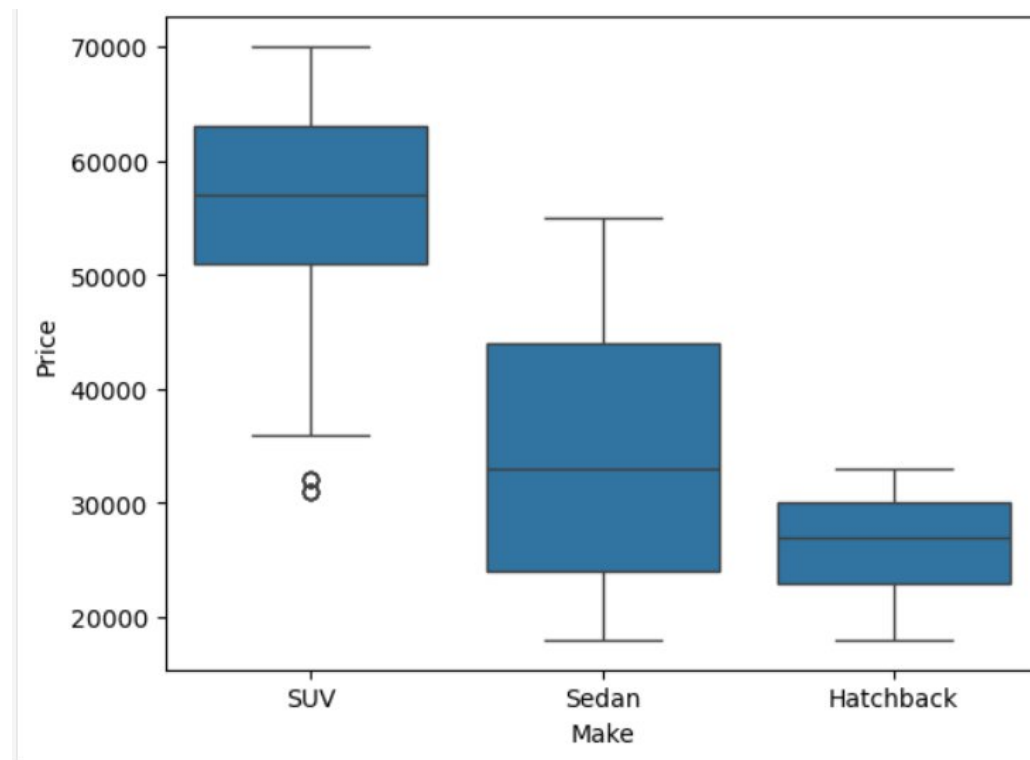
Relationship between Age and Price



Observation

- Strong Positive Correlation: For both genders, there is a clear, strong positive correlation between Age and Price
- After approximately age 50, the Price appears to level off or slightly decrease.
- The price lines for both genders are relatively close, but Female prices are more volatile
- Prices for younger Females show higher initial volatility, peaking higher and dipping lower than males in their early 20s, before stabilizing into the common upward trend.

Relationship between Price and Make((SUV, Sedan, Hatchback))



Observation

1. SUV

- Median (Q2): Highest, approximately 57,000.
- Interquartile Range (IQR): Moderately large, spanning from roughly 52,000 (Q1) to 64,000 (Q3).
- Range: Very wide, with the upper whisker reaching about 70,000.
- Outliers: There is one clear outlier marked below the lower whisker, around 31,000.
- Overall: The SUV category has the highest prices overall.

2. Sedan

- Median (Q2): Intermediate, approximately 33,000.
- Interquartile Range (IQR): Very wide, spanning from roughly 24,000 (Q1) to 44,000 (Q3). This is the widest IQR, indicating the greatest price variability in the middle 50% of the data.

- Range: Wide, extending from about 18,000 to 55,000.

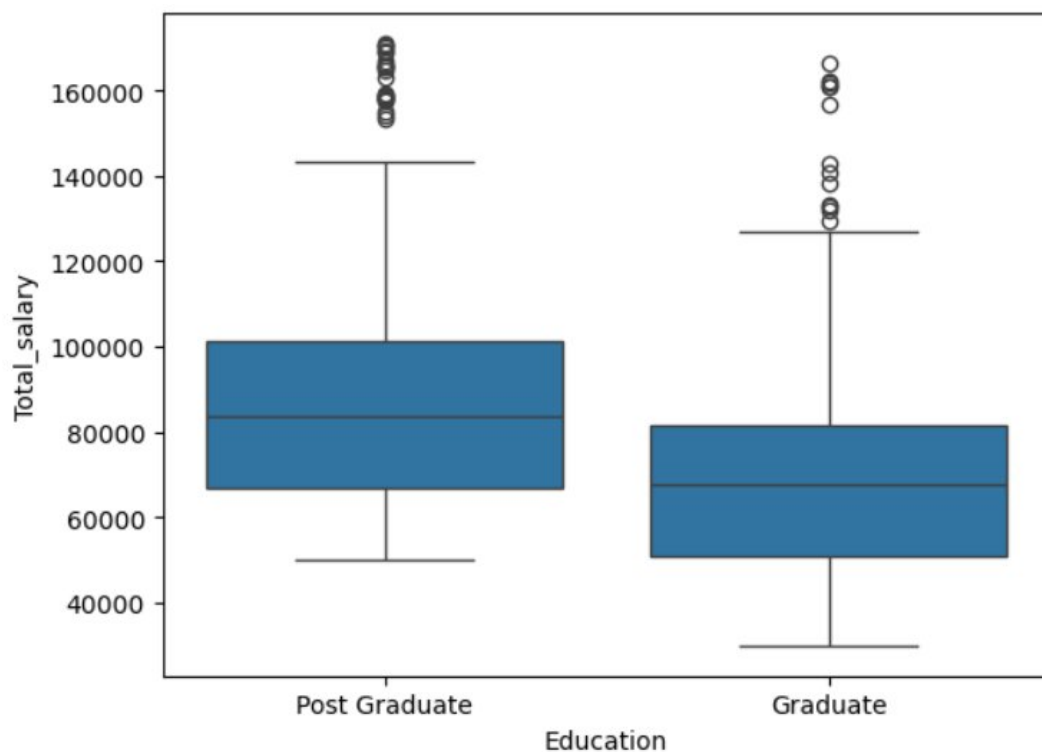
3. Hatchback

- Median (Q2): Lowest, approximately 27,000.
- Interquartile Range (IQR): Narrowest, spanning from roughly 24,000 (Q1) to 30,000 (Q3). This indicates the least price variability and highest concentration of prices.
- Range: The smallest range, extending from about 18,000 to 33,000.
- Overall: The Hatchback category has the lowest prices overall.

A clear price hierarchy established in the dataset: SUV > Sedan > Hatchback.

Sedans exhibit the largest price spread for the middle 50% of their data, suggesting Sedans encompass a broader range of models or features within their typical pricing segment.

Relationship between Total salary and Education



Observation

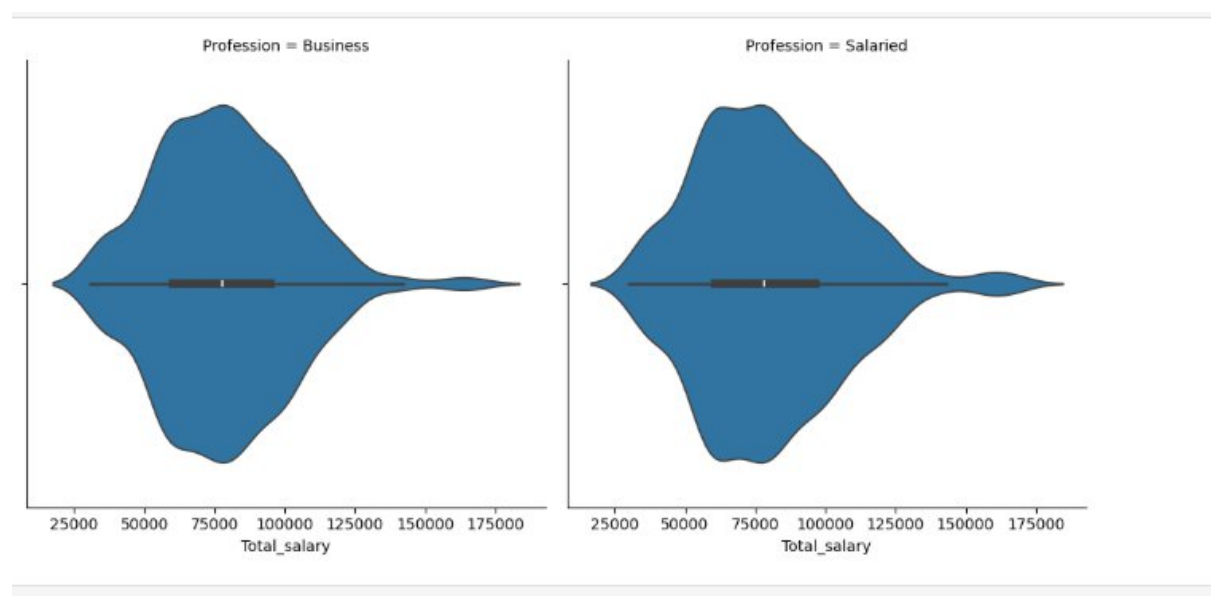
1. Post Graduate

- Median (Q2): Higher, approximately 84,000.
- Interquartile Range (IQR): Spans from roughly 68,000 (Q1) to 102,000 (Q3). This is an IQR of about 34,000.
- Whiskers: The lower whisker reaches about 50,000. The upper whisker reaches about 145,000.
- Outliers: Several clear outliers are present above the upper whisker, ranging from about 150,000 to over 160,000.

2. Graduate

- Median (Q2): Lower, approximately 68,000.
- Interquartile Range (IQR): Spans from roughly \$52,000 (Q1) to 82,000 (Q3). This is an IQR of about 30,000.
- Whiskers: The lower whisker reaches about \$30,000. The upper whisker reaches about 128,000.
- Outliers: Several clear outliers are present above the upper whisker, ranging from about 130,000 to over 160,000.

Relationship between Total salary and Profession



Business Profession

- Central Tendency (Median): The median is around 75,000 to 80,000. The central box (IQR) is relatively narrow.
- Density/Spread (Violin Shape): The plot is bimodal, showing two main peaks in density.
- A significant concentration (highest width) is around 50,000 to 80,000.
- A second, smaller concentration (second peak) is visible around 100,000 to 125,000.
- Range: The tails extend to a similar maximum as the Salaried group, up to about 180,000

Salaried Profession

- Central Tendency (Median): The median is very similar to the Business group, also around 75,000 to 80,000. The central box (IQR) is also relatively narrow.
- Density/Spread (Violin Shape): The plot is unimodal, showing one primary peak in density.
- The vast majority of the density is concentrated in a single, large peak around 75,000 to 95,000.
- Range: The range is similar to the Business group, extending up to about 180,000.

Correlation: visualizes the correlation coefficients between pairs of different variables in a dataset.



Variables: The variables being analyzed are:

- Age
- No_of_Dependents
- Salary
- Partner_salary
- Total_salary
- Price

Scale (Color Bar): The color bar on the right indicates the strength and direction of the correlation:

- Strong Positive Correlation (close to +1.0): Dark blue/purple colors (e.g., 0.80 between Age and Price). This means as one variable increases, the other tends to increase.
- No Correlation (close to 0.0): Yellow/Light green colors (e.g., 0.087 between Salary and Partner_salary). This means there is no linear relationship between the variables.
- Strong Negative Correlation (close to -1.0): Dark red colors (though none are extremely negative, the -0.19 between Age and No_of_Dependents is the most negative value shown). This means as one variable increases, the other tends to decrease.
- Perfect Correlation (Diagonal): The main diagonal of the matrix always shows a correlation of 1 (e.g., Age with Age, Salary with Salary), as a variable is perfectly correlated with itself.
- Insights from the Correlations:

- Total_salary is highly and positively correlated with Partner_salary (0.82) and Salary (0.64), which is expected since total salary is likely a sum or combination of the individual and partner salaries.
- Price has a very strong positive correlation with Age (0.80), suggesting that older individuals in this dataset tend to be associated with higher prices.
- No_of_Dependents shows generally weak correlations with most other variables, particularly with Salary (−0.032) and Partner_salary (0.14).
- Age and No_of_Dependents show a weak negative correlation (−0.19).

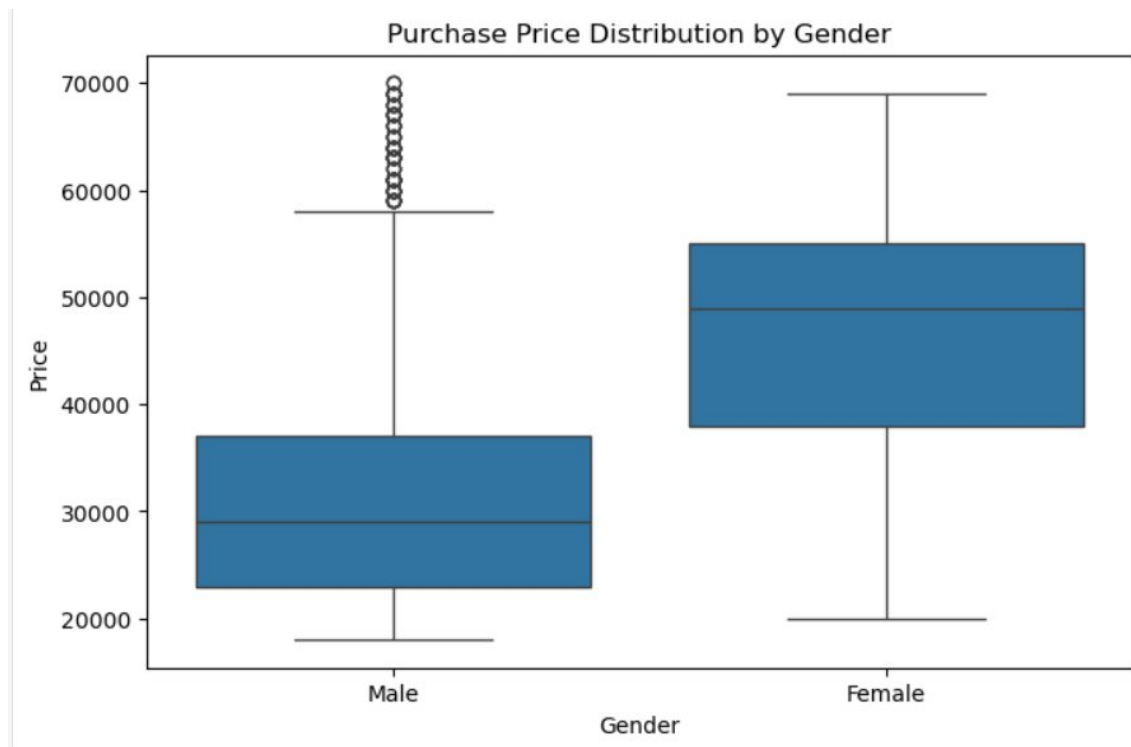
#1. Do men tend to prefer SUVs more compared to women?

Gender	Hatchback	SUV	Sedan
Female	4.559271	52.583587	42.857143
Male	45.28754	9.904153	44.808307



- Based on the clustered bar chart, the answer is No, men do not tend to prefer SUVs more compared to women in this dataset.
- Male SUV Count: The count of Male purchases for SUV is approximately 125.
- Female SUV Count: The count of Female purchases for SUV is approximately 175.
- Conclusion: Since the count of Female purchases is higher than the count of Male purchases for SUVs, women in this dataset tend to prefer SUVs more than men do.

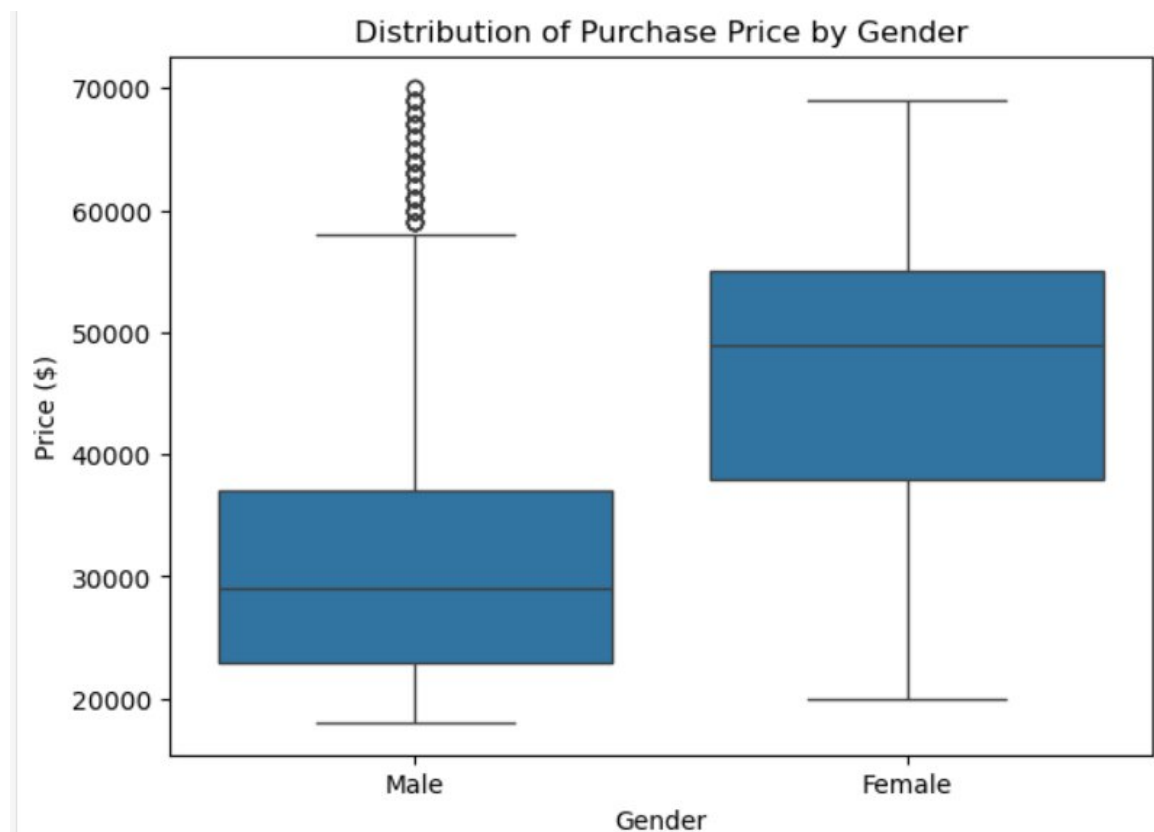
#3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?



- A salaried male is an easier target for an SUV sale over a Sedan sale.
- The provided chart, "Purchase Price Distribution by Gender," offers absolutely no evidence or data to support or refute Sheldon Cooper's specific claim.
- The box plot only shows that:
- The median purchase price is significantly higher for Females (~47,000) than for Males (~29,000).
- The spread of prices (IQR) is also higher for Females.
- Males have a greater number of high-priced outliers.
- To test Sheldon's claim, you would need a visualization that combines Gender, Profession (Salaried vs. Other), and Vehicle Make (SUV vs. Sedan), likely comparing the count or proportion of purchases.

#4. How does the the amount spent on purchasing automobiles vary by gender?

Gender	Mean	Median	count
Female	47705.17	49000	329
Male	32416.13	29000	1252



"Distribution of Purchase Price by Gender," the amount spent on purchasing automobiles varies significantly by gender, with females spending substantially more than males.

#5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

Total Money Spent on Automobiles Grouped by Personal Loan Status:

Personal_loan	Price
NO	28990000
Yes	27290000

#6. How does having a working partner influence the purchase of higher-priced cars?

Partner working	Mean	Median	count
NO	36000	31000	713
Yes	35267.28	31000	868



- **Median Price is Identical:** The typical purchase price is exactly the same for both groups, with the median price being approximately \$31,000. This suggests that having a working partner does not significantly raise or lower the price of the average vehicle purchased.
- **Maximum Price Potential:** Both groups have the same minimum and maximum observed prices, suggesting that the highest- and lowest-price purchases occur regardless of the partner's working status.